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Tactile guidance system for string instruments

Abstract

A tactile guidance system for string instruments is used to teach correct finger position on a string of a string instrument to produce a desired musical note with correct pitch and intonation. The system includes an upper band support structure positioned about an upper end of a fingerboard, a lower support structure positioned about a lower end of the fingerboard, and a tactile band secured between the support structures and positioned above an underlying string. The tactile band includes a tactile indicator that is depressible by a finger of a user causing depression of the tactile band and causing depression of the underlying string against the fingerboard at a predetermined musical note position. The system can also include a bow barrier coupled to the lower band support structure to assist in positioning a bow used to play the instrument at a desired angle.

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS (1) I hereby claim benefit under Title 35, United States Code, Section 119(e) of United States provisional patent application Ser. No. 63/713,760 filed Oct. 30, 2024. The 63/713,760 application is currently pending. The 63/713,760 application is hereby incorporated by reference into this application.

STATEMENT REGARDING FEDERALLY SPONSORED RESEARCH OR DEVELOPMENT
(1) Not applicable to this application.
BACKGROUND
(2) The example embodiments described herein are generally directed to the field of music education and more specifically directed to tactile guidance systems for string instruments to teach correct finger placement on strings of an instrument to produce a desired note.
(3) String instruments present a unique challenge to the beginning musical student in that correct finger position to repeatably achieve a desired note from the instrument is not indicated on the instrument itself but, rather, is only learned through extensive sessions of trial and error. The learning curve to correct finger position proficiency is long, can cause student frustration, and can

even cause a student to give up on learning how to play the instrument. A system that will shorten the learning curve to correct finger position proficiency is needed to encourage continued practice by beginning students.

SUMMARY

(4) Some of the various embodiments of the present disclosure relate to tactile guidance systems for string instruments that will help a beginning student musician to quickly learn correct finger placement on the strings of the instrument to produce a desired note.

(5) A tactile guidance system for string instruments generally includes an upper band support structure positioned about an upper end of a fingerboard of the instrument, a lower support structure positioned about a lower end of the fingerboard, and one or more tactile bands secured between the upper and lower support structures. In its position between the upper and lower support structures, the tactile bands are respectively located above an underlying string of the instrument. The tactile bands include one or more tactile indicators that are depressible by a finger of a user. The depression of the one or more tactile indicators causes depression of the tactile band and, subsequently, causes depression of the underlying string against the fingerboard of the instrument at a predetermined, correct musical note position. In certain embodiments, the tactile guidance system can also include a bow barrier that is coupled to the lower band support structure. The position of the bow barrier atop a body of the instrument is used to assist a student in positioning a bow used to play the instrument at a desired angle.

(6) There has thus been outlined, rather broadly, some of the embodiments of the present disclosure in order that the detailed description thereof may be better understood, and in order that the present contribution to the art may be better appreciated. There are additional embodiments that will be described hereinafter and that will form the subject matter of the claims appended hereto. In this respect, before explaining at least one embodiment in detail, it is to be understood that the various embodiments are not limited in their application to the details of construction or to the arrangements of the components set forth in the following description or illustrated in the drawings. Also, it is to be understood that the phraseology and terminology employed herein are for the purpose of the description and should not be regarded as limiting.

(7) To better understand the nature and advantages of the present disclosure, reference should be made to the following description and the accompanying figures. It is to be understood, however, that each of the figures is provided for the purpose of illustration only and is not intended as a definition of the limits of the scope of the present disclosure. Also, as a general rule, and unless it is evidence to the contrary from the description, where elements in different figures use identical reference numbers, the elements are generally either identical or at least similar in function or purpose.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. **1** is a top view of a tactile guidance system for string instruments as applied to a string instrument in accordance with an example embodiment.

(2) FIG. **2** is a front view of the tactile guidance system for string instruments of FIG. **1**.

(3) FIG. **3** is a side view of the tactile guidance system for string instruments of FIG. **1**.

(4) FIG. **4** is a close-up perspective view of the tactile indicator on a tactile band of the tactile guidance system for string instruments of FIG. **1**.

(5) FIG. **5** is a front perspective view of the tactile guidance system for string instruments of FIG. **1**.

(6) FIG. **6** is a perspective partial view of the tactile guidance system for string instruments of FIG. **1**.

(7) FIG. **7** is a perspective view of a tactile guidance system for string instruments as applied to a string instrument wherein the tactile guidance system additionally includes a bow barrier in accordance with an example embodiment.
(8) FIGS. **8**A, **8**B, and **8**C are perspective views of the bow barrier and a lower band support structure of the tactile guidance system for string instruments of FIG. **7**.
(9) FIG. **9** is a perspective view of neck indicators that can be used in combination with the tactile guidance system for string instruments.
(10) FIG. **10** is a perspective view of a tactile guidance system for string instruments as applied to a string instrument with frets in accordance with an example embodiment.
(11) FIG. **11** is a top perspective view of the tactile guidance system for string instruments of FIG. **7** as applied to a string instrument.
(12) FIG. **12** is a top perspective view of a tactile guidance system for string instruments as applied to a string instrument.

DETAILED DESCRIPTION

A. Overview

(13) Some of the various embodiments of the present disclosure relate to a tactile guidance system for string instruments. The tactile guidance system is used to teach finger position on a string of the instrument to produce a desired musical note with correct pitch and intonation. The tactile guidance system for string instruments reduces the amount of time typically needed to learn correct finger positioning for a string instrument.
(14) The tactile guidance system for string instruments **10** described herein is described in reference to various known elements of a string instrument **12**. These known elements are illustrated in the drawings and include a body **11**, a scroll **13**, a peg box **14**, pegs **15**, a neck **16**, a fingerboard (or fretboard dependent upon the string instrument **12**) **17**, a bridge **18**, and a plurality of strings **19**. The fingerboard **17** includes an upper end **17***a* proximate the peg box **14** that is supported by the neck **16** of the string instrument **12**. The fingerboard **17** also includes a lower end **17***b* that is positioned atop the body **11** of the string instrument **12**. The term “fretboard” is interchangeable as item number **17** with the term “fingerboard” dependent upon the type of string instrument being referenced. String instruments **12** include, but are not limited to, violins, violas, cellos, double basses, guitars, banjos, and ukeleles.
(15) A tactile guidance system for string instruments **10** generally includes an upper band support structure **20**, a lower band support structure **40**, and a tactile band **60**. The upper band support structure **20** is positioned about a plurality of strings **19** and an upper end **17***a* of a fingerboard **17** of the string instrument **12**. The lower band support structure **40** is positioned about the plurality of strings **19** and about a lower end **17***b* of the fingerboard **17** of the string instrument **12**. The tactile band **60** is secured between the upper band support structure **20** and the lower band support structure **40**. The tactile band **60** is additionally suspended above an underlying one of the plurality of strings **19** and includes a tactile indicator **63**. The tactile indicator **63** is individually depressible by a finger of a user thereby causing depression of the tactile band **60** and causing depression of the underlying string **19** against the fingerboard **17** at a predetermined musical note position.
(16) In certain embodiments, the tactile guidance system for string instruments **10** includes a plurality of tactile bands **60** secured between the upper band support structure **20** and the lower band support structure **40** where the number of tactile bands can match or be less than the number of strings **19** on the string instrument **12**.
(17) In certain embodiments, the tactile band **60** includes a plurality of tactile indicators where at least a portion of the tactile indicators **63** represent a corresponding plurality of predetermined musical note positions. The plurality of tactile indicators **63** may be of the same or different shapes/textures and can also include a visual indication such as a color or shape.
(18) In certain embodiments, a first end of the tactile band **60** is secured at a notch of the upper band support structure **20** while the second end of the tactile band **60** is secured at a notch of the

lower support structure **40**. In other embodiments, the first end of the tactile band **60** is secured at the upper band structure **20** with a first band clamp **29** while the second end of the tactile band **60** is secured at the lower and structure with a second band clamp **49**. The first band clamp **29** and the second band clamp **49** may be magnetically couplable to the upper band support structure **20** and the lower band support structure **40**, respectively.

(19) In certain embodiments, the tactile guidance system for string instruments **10** additionally includes a bow barrier **70**. The bow barrier **70** is coupled to the lower band support structure **40** in a position above the body **11** of the string instrument **12** and assists a user in positioning a bow used to play the string instrument **12** at a predetermined angle. The bow barrier **70** can include a second portion **81** that, along with the bow barrier **70** itself, defines a channel through which a bow used to play the string instrument **12** can be drawn.

B. Upper Band Support Structure

(20) FIGS. **1-6** illustrate an example embodiment of the upper band support structure **20** of the tactile guidance system for string instruments **10**, which is preferably of a unitary configuration and is fabricated from a plastic material such as, without limitation, one or more of polypropylene, polyethylene, polystyrene, polyvinyl chloride, polycarbonate, polyethylene terephthalate, and/or synthetic or semi-synthetic materials that can produce the upper band support structure with desired levels of rigidity and flexibility for removably or fixedly securing about (i.e., around or substantially around) a string instrument **12**; rubber, metal, and wood are also possible materials.

(21) The upper band support structure **20** of FIGS. **1-6** has an upper cross portion **21** in a substantially horizontal orientation, with an arched or flat upper surface corresponding to the shape of the fingerboard/fretboard **17**, that is supported by a first side member **23** and a second side member **24**, each of which is oriented substantially perpendicularly to the upper cross portion **21**. A first wrap portion extends outward from a lower end of the first side member **23** in a horizontal orientation toward a second wrap portion **26** that extends in a horizontal orientation from the second side member **24**; the first wrap portion is not shown but is the same as the second wrap portion **26** in an opposing orientation.

(22) Notably a gap is provided between the first wrap portion and the second wrap portion **26** such that the first side member **23** and the second side member **24** can be flexed outward and away from one another to accommodate a width of a neck **16** of a string instrument **12**, such as the violin shown in the Figures. The flexing of the first side member **23** and the second side member **24** enables the upper band support structure **20** to be positioned at the upper end **17***a* of the fingerboard **17** about the neck **16** of the string instrument **12** with the first wrap portion and the second wrap portion **26** returning from flexing to a position below and in contact with the underside of the neck **16**; the first side member **23** and the second member **24** return to positions immediately proximate the sides of the neck **16** and the upper cross portion **21** is placed in a position above the neck **16** and above the strings **19** of the string instrument **12**. The final location of the upper band support structure is near the peg box **14** in a position relative to the string instrument **12** that enables desired finger placement on the fingerboard **17** for producing a desired musical note.

(23) In FIGS. **1-6**, the upper cross portion **21** of the upper band support structure **20** is provided with one or more notches **28** about which an end of the tactile band **60** can be wrapped and secured within. In other embodiments, see, for example, FIG. **7**, the notches are eliminated and the upper cross portion **21** is provided with an un-notched surface about which the end of the tactile band **60** is wrapped. A band clamp **29**, seen in FIGS. **7** and **11**, can be used in conjunction with the notched or un-notched upper cross portion **21** to further secure the end of the tactile band **60** at the upper band support structure **20**. The illustrated band clamp **29** is generally an inverted u-shaped plastic component that slides over the end of the tactile band **60** in an interference fit allowing for removal and/or replacement of the band clamp **29**. The band clamp **29** may include a tacky or textured surface to enhance its grip about the end of the tactile band **60** and upper cross portion **21**. The

upper band support structure **20** itself may also include a tacky or textured surface to enhance its grip about the fingerboard **17**.

(24) FIGS. **7**, **10**, and **11** illustrate an alternative configuration of the upper band support structure **20** wherein the first wrap portion and the second wrap portion **26** are replaced with a respective first side clamp **30** and a second side clamp **31**, each of which is a two-wall configuration having an upper wall **32** and a side wall **33**. The side wall **33** forms a right angle with the upper wall **32** to define a right angle cavity **34** whereby the upper walls **32** are positionable atop the upper end of the fingerboard **17** of the string instrument **12** and the side walls **33** are positionable immediately proximate the neck **16** of the string instrument upon return from outward flexing of the first side member **23** and the second side member **24**; return from outward flexing positions the upper band support structure **20** about the string instrument **12**. In certain embodiments, one or both of the first side clamp **30** and the second side clamp **31** are equipped with a tab **35**, which is finger-grabbable, to facilitate the outward flexing of the first side member **23** and/or second side member **24**.

(25) It should be noted that, in the instance that the string instrument **12** is an instrument that includes a fretboard (e.g., such as a guitar, banjo, mandolin, ukelele, etc.) rather than a fretless fingerboard **17**, the upper walls **32** of the first side clamp **30** and the second side clamp **31** are positionable atop the fretboard as shown in FIG. **10**. It should also be noted, that in certain embodiments, one or both of the first side clamp **30** and second side clamp **31** may include a third wall similar to the configuration of the lower band support structure **40** described herein to create a u-shaped cavity to provide a three-sided grip about the fingerboard **17** (or fretboard).

(26) FIG. **12** illustrates another alternative configuration of the upper band support structure **20** wherein the upper walls **32** of each of the first side clamp **30** and the second side clamp **31** simply rest atop the upper end **17***a* of the fingerboard **17** and the side walls **33** of each of the first side clamp **30** and the second side clamp **31** rest proximate the neck **16** of the string instrument **12**. In this configuration, an elastic band **37** extends from the tab **35** on the first side clamp **30** underneath the neck **16** of the string instrument **12** to the tab **35** of the second side clamp **31** with the tension of the elastic band **37** functioning to removably secure the upper band support structure **20** to the string instrument **12**. In this configuration, the upper band support structure **20** additionally includes a ferromagnetic material (not shown) in the upper cross portion **21** to interface with a magnet within each of the band clamps **49**; further detail regarding the ferromagnetic material and magnetic band clamps are further described herein below.

C. Lower Band Support Structure

(27) The exemplary lower band support structure **40** of the tactile guidance system for string instruments **10** depicted in FIGS. **1-11** is similar to the upper band support structure **20** in that the lower band support structure **40** is preferably of a unitary configuration and is fabricated from a plastic material such as, without limitation, one or more of polypropylene, polyethylene, polystyrene, polyvinyl chloride, polycarbonate, polyethylene terephthalate, and/or synthetic or semi-synthetic materials that can produce the lower band support structure **40** with desired levels of rigidity and flexibility for removably or fixedly securing about a string instrument **12**; rubber, metal, and wood are also possible materials.

(28) Further, the lower band support structure **40** also includes an upper cross portion **41** in a substantially horizontal orientation, having an arched or flat upper surface corresponding to the shape of the fingerboard/fretboard **17**, that is supported by a first side member **43** and a second side member **44**, each of which is oriented substantially perpendicularly to the upper cross portion **41**. However, with the lower band support structure **40**, each of the first side member **43** and the second side member **44** extend to a respective first side clamp **50** and a second side clamp **51**, each of which includes an upper wall **52**, side wall **53**, and lower wall **54** that join together to form a u-shaped cavity **55**.

(29) The first side member **43** and the second side member **44** of the lower band support structure **40** can be flexed outward to accommodate the placement of the lower support structure **40** about a

lower end **17***b* of the fingerboard **17** of the string instrument **12**. The return of the first side member **43** and the second side member **44** from the outward flex results with the sides of the lower end of the fingerboard **17** positioned within the u-shaped cavity **55** of the first side clamp **50** and the second side clamp **51**. The u-shaped cavity **55** then presents the upper wall of the **52** of the first side clamp **50** and the second side clamp atop the fingerboard **17**, the side wall **53** of the first side clamp **50** and the second side clamp **51** immediately proximate the side/edge of the fingerboard **17**, and the lower wall of the first side clamp **50** and the second side clamp **51** immediately beneath the fingerboard **17**; the upper cross portion **41** of the lower band support structure **40** is presented in a position above the strings **19** of the string instrument **12**. In certain embodiments, each of the first side clamp **50** and the second side clamp **51** are equipped with a tab **56**, which is finger-grabbable, to facilitate the outward flexing of the first side member **43** and second side member **44**.
(30) In FIGS. **1-6**, as with the upper band support structure **20**, the upper cross portion of the lower band support structure **40** is provided with one or more notches **48** about which an end of the tactile band **60** can be wrapped and secured within. In other embodiments, see, for example, FIG. **7**, the notches are eliminated and the upper cross portion **41** is provided with an un-notched surface about which the end of the tactile band **60** is wrapped. A band clamp **49**, seen in FIGS. **7** and **11**, can be used in conjunction with the notched or un-notched upper cross portion **41** to further secure the end of the tactile band **60** at the lower band support structure **40** in a position above the string **19** of the string instrument **12**. The illustrated band clamp **49** is generally an inverted u-shaped plastic component that slides over the end of the tactile band **60** in an interference fit allowing for removal and/or replacement of the band clamp **49**. The band clamp **49** may include a tacky or textured surface to enhance its grip about the end of the tactile band **60** and upper cross portion **41**. The lower band support structure **40** itself may also include a tacky or textured surface to enhance its grip about the fingerboard **17**.
(31) In certain embodiments, each of the first side clamp **50** and the second side clamp **51** of the lower band support structure **40** includes a projection **57** at a rear of the side wall **53** serving as an interface for aligning and attaching a bow barrier **70** (depicted in FIGS. **7-9** and **11-12** and described further herein) to the lower band support structure **40**.
(32) It should be noted that, in the instance that the string instrument **12** is an instrument that includes a fretboard (e.g., such as a guitar, banjo, mandolin, ukelele, etc.) rather than a fretless fingerboard **17**, the upper walls **52** of the first side clamp **50** and the second side clamp **51** are positionable atop the fretboard as shown in FIG. **10**.
(33) FIG. **12** illustrates another alternative configuration of the lower band support structure **40** wherein the upper walls **52** of each of the first side clamp **50** and the second side clamp **51** simply rest atop the lower end **17***b* of the fingerboard **17** and the side walls **53** of each of the first side clamp **50** and the second side clamp **51** rest proximate the side edge of the fingerboard **17** of the string instrument **12**. In this configuration, an elastic band **59** extends from the tab **56** on the first side clamp **50** underneath the lower end **17***b* of the fingerboard **17** of the string instrument **12** to the tab **56** of the second side clamp **51** with the tension of the elastic band **59** functioning to removably secure the lower band support structure **40** to the string instrument **12**. In this configuration, the lower band support structure **40** additionally includes a ferromagnetic material **58** in the upper cross portion **41** to interface with a magnet within each of the band clamps **49**; further detail regarding the magnetic band clamps are further described herein below.
D. Tactile Band
(34) One or more tactile bands **60**, e.g., four tactile bands **60** for a four-stringed instrument or six tactile bands for a six-stringed instrument, are included in the tactile guidance system for string instruments **10**. Each of the tactile bands **60** is preferably made of a flexible and/or stretchable material that is capable of flexing and/or stretching then returning to its original unflexed/unstretched configuration. The flexible and/or stretchable material can comprise one or more, without limitation, elastomers such as natural rubber, synthetic rubber, silicone, and

polyurethane.
(35) Each of the tactile bands **60** is of a length suitable to the instrument to which it is being applied that allows the tactile bands **60** to extend between the upper band support structure **20** and the lower band support structure **40**. Sufficient additional length is provided at each of the upper end **61** and the lower end **62** of the tactile band **60** so as to enable the upper end **61** and the lower end **62** of the tactile band **60** to engage and/or wrap about the upper band support structure **20** and the lower band support structure **40** (in either of a notched or unnotched configuration). If desired and/or necessary, a band clamp **29** can be placed about the engaged tactile band **60** at the upper band support structure **20** and lower band support structure **40** to more firmly secure the respective upper end **61** and the lower end **62** of the tactile band **60**.
(36) Each of the tactile bands **60** is preferably provided with one or more tactile indicators **63** that indicate to the user the correct positioning of a finger to obtain a desired predetermined note upon the user depressing, with their finger, the tactile indicator **63**. Depression of the tactile indicator **63** causes depression of the tactile band **60** and depression of the underlying string **19** against the fingerboard **17** of the string instrument **12**. In certain embodiments, the tactile band **60** is provided with a plurality of tactile indicators **63** for finger positioning of a corresponding plurality of predetermined notes (a singular tactile indicator **63** on the tactile band **60** can be depressed or a plurality of tactile indicators **63** on the tactile band can be depressed simultaneously). In certain embodiments, each of the plurality of tactile indicators **63** on the tactile band **60** is different from the other of the plurality of tactile indicators **63** on the tactile band **60** while in other embodiments the tactile indicators **63** are the same at different locations along the tactile band **60**. In certain embodiments, each of a plurality of the tactile bands **60** are configured with one or more tactile indicators **63** that are different from the tactile indicators **63** found on another of the plurality of tactile bands **60** for a particular instrument.
(37) The one or more tactile indicators **63** on the respective tactile band **60** may be any kind of shape or texture that rises above the upper surface **64** of the tactile band **60**. In a preferred embodiment, the tactile indicators **63** are unitary with the tactile band **60** while in other embodiments the tactile indicators **63** are positioned atop the tactile band **60** after manufacture of the tactile band **60** where the tactile indicators **63** are glued or otherwise attached to the tactile band. In certain embodiments, the one or more tactile indicators **63** may additionally be color-coded for additional visual recognition beyond tactile recognition by the user. Because correct positioning of the tactile indicator **63** above the underlying string **12** is key to finger placement and production of a desired note, it is preferable that a skilled musician with knowledge in playing the instrument **12** to which the tactile band **60** is being applied use their knowledge for the tactile band **60** and tactile indicator **63** positioning. In certain embodiments, to assist in correct placement of the tactile band **60** and the tactile indicators **63**, one or more location indicators **65** (e.g., colored tape or the like) may be placed directly on the fingerboard **17** to indicate a location for placement of a tactile indicator **63**; see, for example, FIG. **9**.
(38) FIG. **12** illustrates an alternative configuration wherein the upper end **61** and the lower end **62** of the tactile band **60** are wrapped about an interior surface (not shown) of a respective band clamp **29**, **49** and covered with an outer shell **67** that conceals the end **61**, **62** of the respective tactile band **60**. The outer shell **67** additionally serves to conceal a magnet **68** within the respective band clamp **29**, **49**; the magnet **68** magnetically interfaces with the ferromagnetic material **58** in the upper band support structure **20** or lower band support structure **40** to hold the tactile band **60** in a correct position, with respect to the tactile indicators **63** of the tactile band **60**, relative to a respective one of the strings **19** of the string instrument **12**.
E. Bow Barrier
(39) In certain embodiments, the tactile guidance system for string instruments **10** can additionally include a bow barrier **70** that is positionable behind the lower band support structure **40**. The bow barrier **70** assists a user in positioning a bow at a correct location on the strings **19** of the instrument

12 with that correct location being between the bow barrier **70** and the bridge **18** of the instrument **12**. Further, the bow barrier **70** assists a user in drawing a straight bow with the appropriate inclination (i.e., the inclination or angle of the bow is constrained by the bow barrier **70** to a position where the bow is slightly tilted away from the bridge **18** towards the scroll **13**). FIGS. **7**, **8**A-**8**C, and **11** illustrate an example embodiment of the bow barrier **70**.

(40) The bow barrier **70** is configured similarly to the upper band support structure **20** and the lower band support structure **40** in that the bow barrier is preferably of a unitary configuration and is fabricated from a plastic material such as, without limitation, one or more of polypropylene, polyethylene, polystyrene, polyvinyl chloride, polycarbonate, polyethylene terephthalate, and/or synthetic or semi-synthetic materials that can produce the bow barrier **70** with desired levels of rigidity and flexibility that it might be removably secured to the lower band support structure **40**; rubber, metal, and wood are also possible materials.

(41) The bow barrier **70** includes an upper cross portion **71** in a substantially horizontal orientation that has an arched or flat upper surface corresponding to the shape of the fingerboard/fretboard **17**. The upper cross portion **71** of the bow barrier is supported by a first side member **72** and a second side member **73**, each of which is oriented substantially perpendicularly to the upper cross portion **71**. Each of the first side member **72** and the second side member **73** extend to a respective first base member **74** and a second base member **75**. Each of the first base member **74** and second base member **75** includes a recessed front surface **76** proximate an end portion **77** that includes a slot **78** to receive the projection **57** of the lower band support member **40** and secure the bow barrier **70** to the lower band support structure **40**. In certain embodiments, each of the first base member **74** and the second side base member **75** are equipped with a tab **77**, which is finger-grabbable, to facilitate the outward flexing of the first side member **72** and second side member **73** to assist in positioning the bow barrier **70** about the lower band support member **40**.

(42) Referring to FIG. **12** and an alternative configuration of the bow barrier **70**, the first side member **72** and the second side member **73** of the bow barrier **70** rest atop the fingerboard **17** while an elastic band **80** extends from the tab **77** of the first side base member **74** underneath the fingerboard **17** to the tab **77** of the second side base member **75** to removably secure the bow barrier **70** to the fingerboard **17** of the string instrument **12**. The bow barrier **70** of FIG. **12** additionally includes a second portion **81**, similarly configured to the bow barrier **70**, with an upper cross portion **82** substantially parallel to the upper cross portion **71** of the bow barrier **70**. A first side member **83** and a second side member **84** support the upper cross portion **82**. Each of the first side member **83** and the second side member **84** are permanently or removably secured to the bow barrier **70** by cross-arm **85**. The bow barrier **70** in combination with the second portion **81** of the bow barrier **70** serve to define a channel at the appropriate position on the string instrument **12** through which a bow used to the play string instrument can be correctly positioned and drawn.

F. Operation of Preferred Embodiment

(43) In use, a string instrument **12** is equipped with the tactile guidance system for string instruments **10** by first positioning the upper band support structure **20** and the lower band support structure **40** about the neck **16** and fingerboard **17** of the string instrument **12**.

(44) In placing the upper band support structure **20**, the first side member **23** and the second side member **24** are flexed outward to accommodate the width of the neck **16** of the string instrument so as to allow the first wrap portion and the second wrap portion **26** to slip immediately under the neck **16** proximate the peg box **14**; the flex of the upper band support structure **20** is released to allow the upper band support structure a tight fit about the neck **16**. In an alternative configuration (see FIGS. **7** and **11**), upon the outward flexing of the first side member **23** and the second side member **24**, the first side clamp **30** and the second side clamp **31** of the upper band support structure **20** are positioned with the upper wall **32** of the first side clamp **30** and the second side clamp **31** atop the fingerboard **17** and the side wall **33** immediately proximate the neck **16** in a right angle orientation to the upper wall **32**. In still another configuration, see FIG. **12**, the first side

clamp **30** and the second side clamp **31** rest atop the fingerboard **17** and are removably secured thereto with the elastic band **37** that extends under the neck **16** of the string instrument **12** coupling the tab **35** of the first side clamp **30** to the tab **35** of the second side clamp **31**.

(45) In placing the lower band support structure **40**, the first side member **43** and the second side member are flexed outward to accommodate the width of the fingerboard **17**, e.g., the broadest width of the fingerboard **17**, at the lower end of the fingerboard **17**. The flex of the lower band support structure **40** is released allowing the u-shaped cavity **55** of the first side clamp **50** and the second side clamp **51** to fit tightly about fingerboard **17** with the upper wall **52** of each of the clamps resting on the fingerboard **17**, the side wall **53** of the each of clamps positioned immediately to the side of the fingerboard **17**, and lower wall **54** of each of the clamps positioned immediately underneath the fingerboard **17**. In still another configuration, see FIG. **12**, the first side clamp **50** and the second side clamp **51** rest atop the fingerboard **17** and are removably secured thereto with the elastic band **59** that extends under the fingerboard **17** of the string instrument **12** coupling the tab **56** of the first side clamp **50** to the tab **56** of the second side clamp **51**.

(46) One or more of the tactile bands **60** is then stretched tautly between the upper band support structure **20** and the lower band support structure **40** with the upper end **61** of the tactile band **60** wrapped about and secured to the upper cross portion **21** of the upper band support structure **20**; securing of the upper end **61** of the tactile band is achieved through use of one of the notches in the upper cross portion **21** and/or one of the band clamps **29** positioned about the wrapped upper end **61** of the tactile band **60** and press fit about the upper cross portion. The lower end **62** of the tactile band **60** is similarly secured about the lower band support structure **40**. This procedure can be completed for each of the tactile bands **60** that is desired to be used with the string instrument. Notably, in certain embodiments, each of the upper band support structure **20** and the lower band support structure **40** is preferably configured to accommodate as many tactile bands **60** as there are strings on the string instrument **12**. In an alternative embodiment (see FIG. **12**), the upper end **61** of the tactile band **60** is wrapped about the band clamp **29** itself and the lower end **62** of the tactile band **60** is wrapped about the band clamp **49** itself wherein the magnet **68** within the respective band clamp **29**,**49** interfaces with the ferromagnetic material **58** in the upper band support structure **20** and the lower band support structure **40**, respectively, to hold the tactile band **60** in a desired position.

(47) In placing the tactile band **60** relative to the upper band support structure **20** and the lower band support structure **40**, care is given to the positioning the one or more tactile indicators **63** on the tactile band **60** at a location relative to the fingerboard **17** such that when the tactile indicator **63** of the tactile band **60** and the underlying string **19** of the string instrument **12** are depressed by a user's finger a desired and accurate note (e.g., accurate pitch and intonation) will be produced when drawing a bow across the respective string **19** at a position near the bridge **18** (or when plucking the string **19**). In the instance of a guitar (or other instrument with a fretboard), a desired and accurate note (e.g., pitch and intonation) will be produced by strumming, picking, or plucking the string **19**.

(48) If desired, the bow barrier **70**, with or without the second portion **81** of the bow barrier **70** (see FIG. **12**), may also be attached to the lower band support structure **40** by flexing the first side member **72** and second side member **73** outward and released so as to enable each of the first base member **74** and the second base member **75** to rest atop the body **11** of the string instrument **12** and so as to enable to enable the slot **78** in each of the first base member **74** and the second base member **75** to receive the projections **57** of the lower band support member **40** and, essentially, removably lock the bow barrier **70** in position relative to the lower band support member **40**.

(49) Each of the upper band support member **20**, lower band support member **40**, tactile bands **60**, and bow barriers **70** are sized specifically to accommodate at least one size of a string instrument (e.g., violins are manufactured in multiple sizes). However, it is possible one or more of the upper band support member **20**, lower band support member **40**, tactile bands **60**, and bow barriers **70** can accommodate more than one size of a string instrument **12** and/or different string instruments **12**

(e.g., a violin and a viola). It is also possible that the upper band support member **20**, lower band support member **40**, and bow barriers **70** can accommodate more than one size of a string instrument **12** and/or different string instruments **12** in conjunction with different sets of tactile strings **60** that are made specific to a string instrument **12**.
(50) In the preferred embodiment, the tactile guidance system for string instruments **10** is removably secured to the string instrument, however, it should be noted that the tactile guidance system for string instruments **10** can be fixedly secured to the instrument such as by an appropriate adhesive.
(51) In certain embodiments, one or more of the upper band support member **20**, lower band support member **40**, and one or more tactile bands **60** are placed on the string instrument **12** in a position such that finger placement on a string **19** not covered by a tactile band is still possible. In certain embodiments, one or more of the tactile indicators **63** may correspond to a specific note or a combination of note patterns along a string **19** or combination of strings **19**.
(52) It should be noted that the tactile guidance system for string instruments **10** is usable to establish muscle memory in user for correct finger position. For example, prior to the suppression of the string **19** and the product of sound with a bow, the placement of a finger according to the tactile indicators **63** connects audiation of a note with the position of the fingers.
(53) While various embodiments have been described above, it should be understood that they have been presented by way of example only allowing for features of one embodiment to be combined features of one or more other embodiments. The descriptions are not intended to limit the scope of the technology to the particular forms set forth herein. To the contrary, the present descriptions are intended to cover such alternatives, modifications, and equivalents as may be included within the spirit and scope of the technology as defined by the appended claims and otherwise appreciated by one of ordinary skill in the art. The various embodiments of the present disclosure may be embodied in other specific forms without departing from the spirit or essential attributes thereof, and it is therefore desired that the various embodiments in the present disclosure be considered in all respects as illustrative and not restrictive. Thus, the breadth and scope of a preferred embodiment should not be limited by any of the above-described exemplary embodiments.
(54) Unless otherwise defined, all technical and scientific terms used herein have the same meaning as commonly understood by one of ordinary skill in the art to which this invention belongs. All patent applications, patents, and printed publications cited herein are incorporated herein by reference in their entireties, except for any definitions, subject matter disclaimers or disavowals, and except to the extent that the incorporated material is inconsistent with the express disclosure herein, in which case the language in this disclosure controls. Any headings utilized within the description are for convenience only and have no legal or limiting effect.

Claims

1. A tactile guidance system for string instruments comprising: an upper band support structure positioned about a plurality of strings of a string instrument and positioned about an upper end of a fingerboard of the string instrument; a lower band support structure positioned about the plurality of strings of the string instrument and positioned about a lower end of the fingerboard of the string instrument; and a tactile band secured between the upper band support structure and the lower band support structure in a position above an underlying one of the plurality of strings, wherein the tactile band includes a tactile indicator that is depressible by a finger of a user causing depression of the tactile band and causing depression of the underlying one of the plurality of strings against the fingerboard at a predetermined musical note position.
2. The tactile guidance system of claim 1, further comprising a plurality of tactile bands secured between the upper band support structure and the lower band support structure.
3. The tactile guidance system of claim 2, wherein the plurality of tactile bands comprises a number

of tactile bands that are equivalent to a number of strings on the string instrument.
4. The tactile guidance system of claim 1, wherein the tactile band includes a plurality of tactile indicators, wherein at least a portion of the plurality of tactile indicators represent a corresponding plurality of predetermined musical note positions.
5. The tactile guidance system of claim 4, wherein one of the portion of the plurality of tactile indicators are of a different shape than another one of the portion of the plurality of tactile indicators.
6. The tactile guidance system of claim 4, wherein one of the portion of the plurality of tactile indicators are of a different texture than another one of the portion of the plurality of tactile indicators.
7. The tactile guidance system of claim 1, wherein the tactile indicator includes a visual indicator in a form of a color or shape.
8. The tactile guidance system of claim 1, wherein a first end of the tactile band is secured at a notch in the upper band support structure and wherein a second end of the tactile band is secured at a notch in the lower band support structure.
9. The tactile guidance system of claim 1, wherein a first end of the tactile band is secured at the upper band support structure with a first band clamp and wherein a second end of the tactile band is secured at the lower band support structure with a second band clamp.
10. The tactile guidance system of claim 9, wherein each of the first band clamp and the second band clamp magnetically couplable to the upper band support structure and the lower band support structure.
11. The tactile guidance system of claim 1, wherein the string instrument comprises one of a violin, a viola, a cello, or a double bass.
12. The tactile guidance system of claim 1, wherein the fingerboard is in a form of a fretboard and wherein the string instrument comprises one of a guitar, banjo, or ukelele.
13. A tactile guidance system for string instruments comprising: an upper band support structure positioned about a plurality of strings of a string instrument and positioned about an upper end of a fingerboard of the string instrument; a lower band support structure positioned about the plurality of strings of the string instrument and positioned about a lower end of the fingerboard of the string instrument; a tactile band secured between the upper band support structure and the lower band support structure in a position above an underlying one of the plurality of strings, wherein the tactile band includes a tactile indicator that is depressible by a finger of a user causing depression of the tactile band and causing depression of the underlying one of the plurality of strings against the fingerboard at a predetermined musical note position; and a bow barrier coupled to the lower band support structure, wherein the bow barrier assists in positioning a bow used to play the string instrument at predetermined angle.
14. The tactile guidance system of claim 13, further comprising a plurality of tactile bands secured between the upper band support structure and the lower band support structure.
15. The tactile guidance system of claim 14, wherein the plurality of tactile bands comprises a number of tactile bands that are equivalent to a number of strings on the string instrument.
16. The tactile guidance system of claim 13, wherein the tactile band includes a plurality of tactile indicators, wherein at least a portion of the plurality of tactile indicators represent a corresponding plurality of predetermined musical note positions.
17. The tactile guidance system of claim 13, wherein the bow barrier includes a second portion oriented parallel to the bow barrier that, along with the bow barrier, defines a channel through which a bow used to play a string instrument can be drawn.
18. The tactile guidance system of claim 13, wherein a first end of the tactile band is secured at the upper band support structure with a first band clamp and wherein a second end of the tactile band is secured at the lower band support structure with a second band clamp.
19. The tactile guidance system of claim 13, wherein the string instrument comprises one of a

violin, a viola, a cello, or a double bass.

20. A tactile guidance system for string instruments comprising: an upper band support structure positioned about a plurality of strings of a string instrument and positioned about an upper end of a fingerboard of the string instrument; a lower band support structure positioned about the plurality of strings of the string instrument and positioned about a lower end of the fingerboard of the string instrument; and a plurality of tactile bands secured between the upper band support structure and the lower band support structure in a position above a respective underlying one of the plurality of strings, wherein at least one of the plurality of tactile bands includes a plurality of tactile indicators that are individually depressible by a finger of a user or simultaneously depressible by a plurality of fingers of the user, wherein depression of at least one of the plurality of tactile indicators causes depression of the at least one of the plurality of tactile bands and causes depression of the underlying one of the plurality of strings against the fingerboard at a predetermined musical note position.